

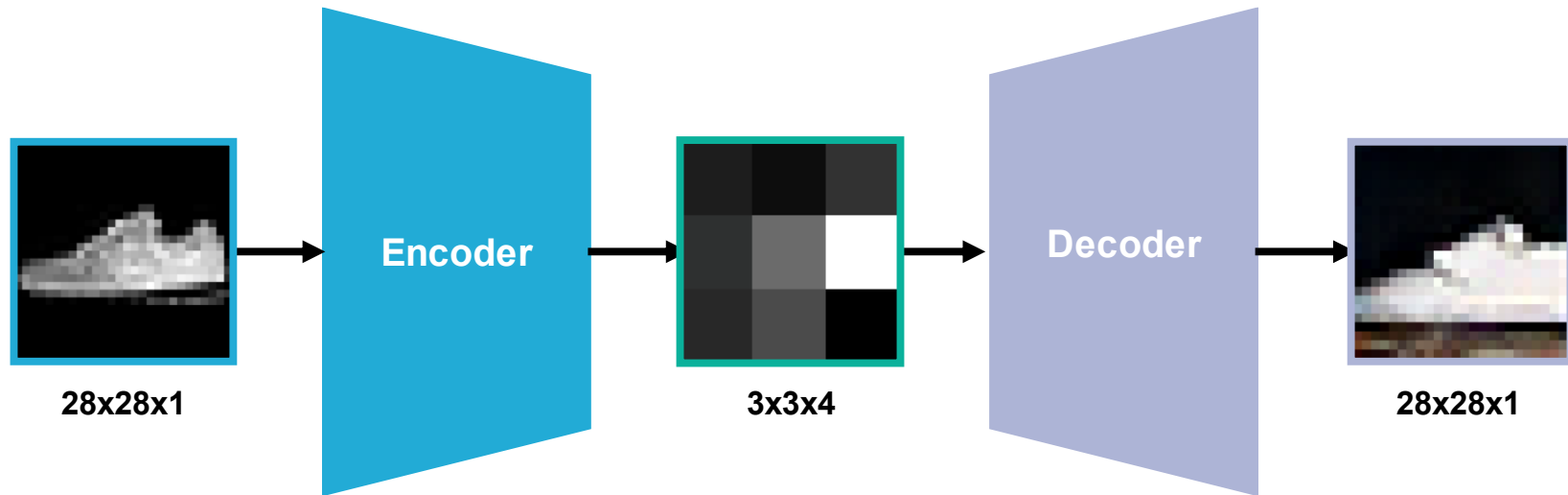
Image generation models

- The Early Reconstruction Foundations (2013–2014)
 - Auto-encoders, VAE
- Adversarial dominance era (2014-2020)
 - GANs, Style GANs,
- Text to Image Diffusion era (2020-2023)
 - Generate images starting from pure Gaussian noise and iteratively removing noise over a series of steps.
 - Stable Diffusion, DALL-E, Midjourney
- The diffusion transformer (2023-2025)
 - Instead of CNNs use ViTs, i.e. as a sequence of visual tokens
- Unified autoregressive & Flow matching systems (2025-Present)
 - GPT-4o, Nano Banana

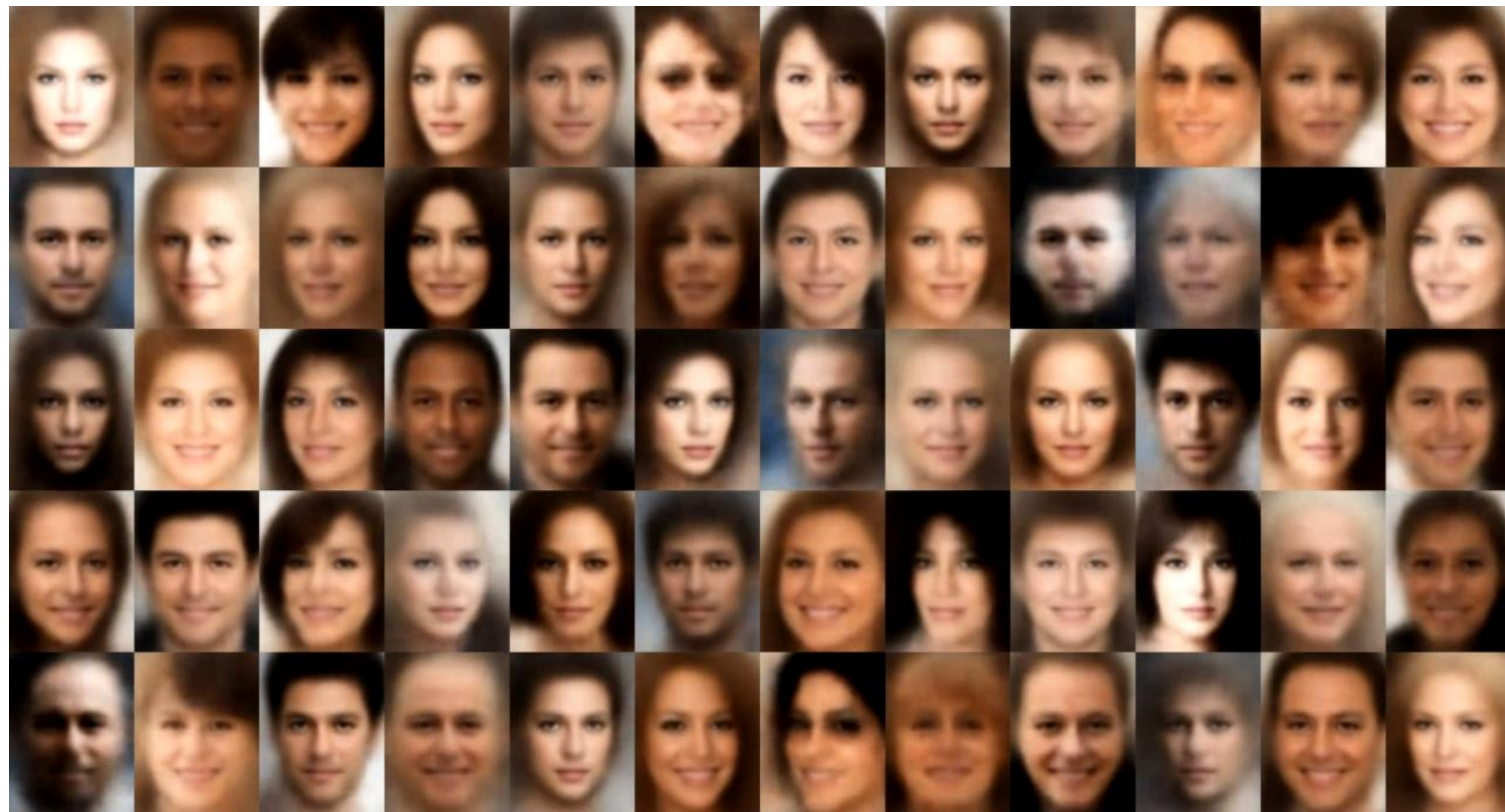
Acknowledgement:

Slides for this lecture derived from the BMDS271 course at Stanford taught by Akshay Chaudari and Tanveer Syeda-Mahmood

VAR-based image generation



Variational Autoencoder

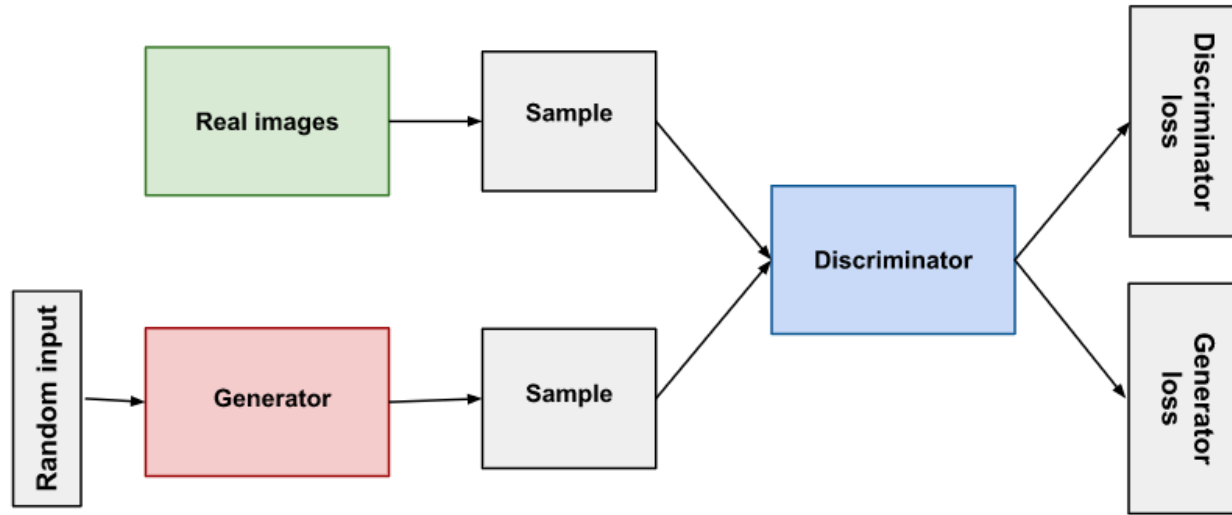


StyleGAN2



Generative Adversarial Network

- Use an additional *discriminator* network to enforce semantic consistency of *generator* images

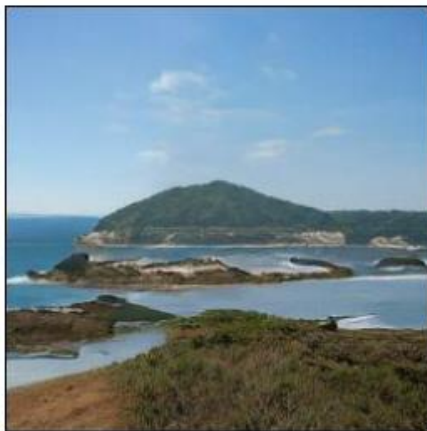


[1] Goodfellow I. et al "Generative Adversarial Nets". NeuRIPS (2014)

[2] Overview of GAN Structure (https://developers.google.com/machine-learning/gan/gan_structure)

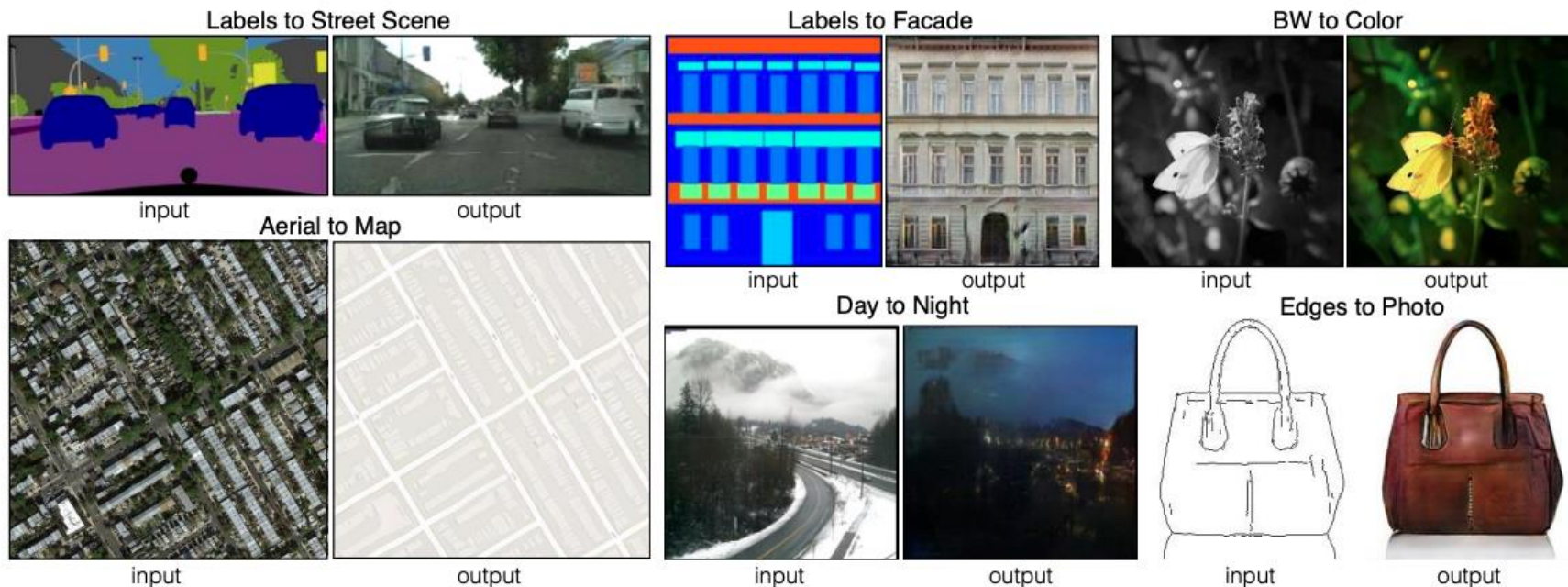
Conditional GANs

- Create generative models conditioned on input features
- Example: Generate an image with a specific class



Conditional GANs

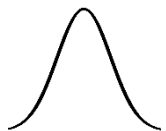
- Create generative models conditioned on segmentation masks



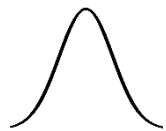
Denoising Diffusion Models

Training: Image $\rightarrow$ Noise

$X(0)$  $X(T)$

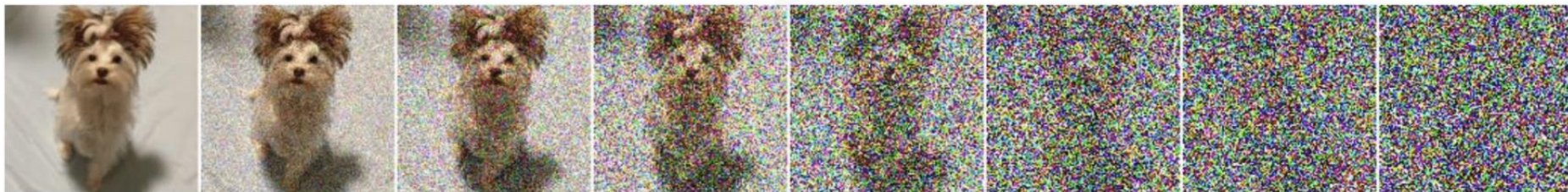


$+ \mathcal{N}_1(0, \sigma^2)$



$+ \mathcal{N}_2(0, \sigma^2)$

...



X^0

X^1

X^2

X^T

$X(0)$  $X(T)$

Inference: Image $\leftarrow$ Noise

Text-to-Image Generative Models

A photo of a corgi wearing a hat in Times Square. It is wearing sunglasses and a beach hat.



A majestic oil painting of a raccoon Queen wearing red French royal gown. The painting is hanging on an ornate wall decorated with wallpaper.

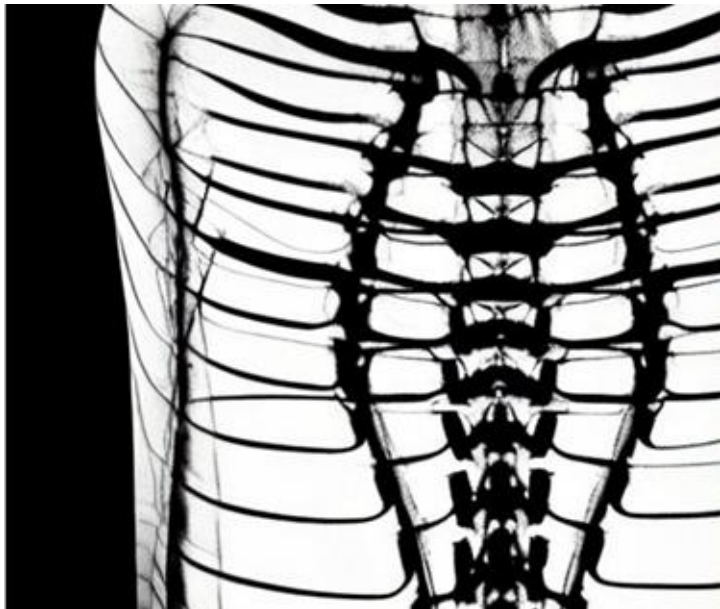


Teddy bears swimming at the Olympics 400m Butterfly event



Text-to-Image Generative Models

A photo of a lung x-ray



A photo of a lung x-ray
with visible pleural effusion



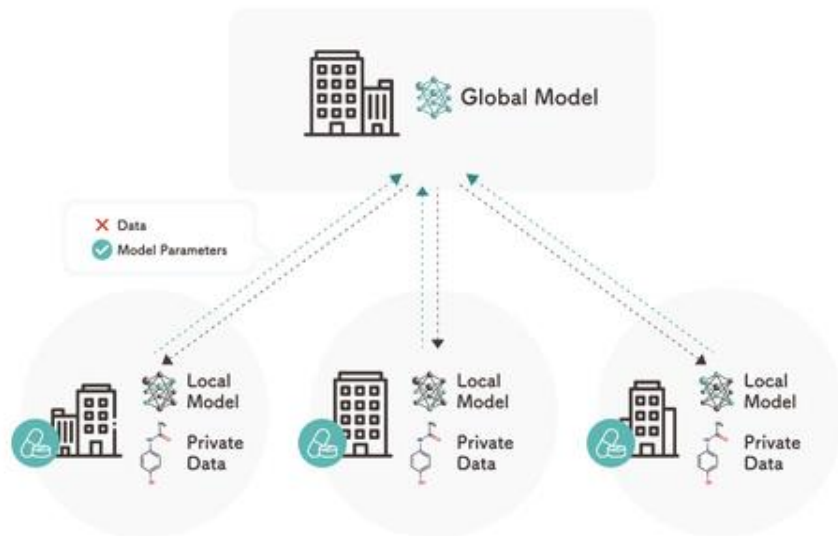
Why Use Synthetic Medical Images?

- Data augmentation
 - Scarcity of data
 - Rare diseases
 - Mitigating Bias



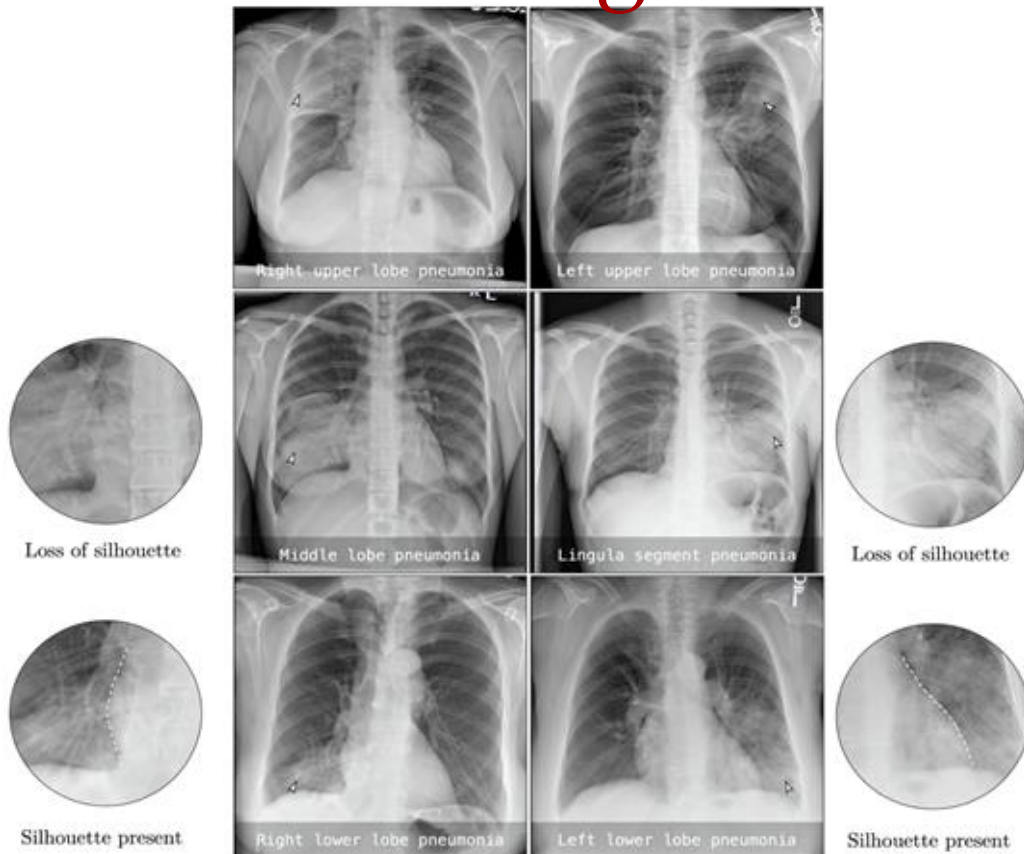
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Why Use Synthetic Medical Images?

- Data augmentation
 - Scarcity of Data
 - Rare diseases
 - Mitigating Bias
- Privacy preservation
- Medical Education





Original Stable Diffusion
"a photo of a lung xray with visible pleural effusion"

"Small right-sided
pleural effusion"



"No pleural effusion"



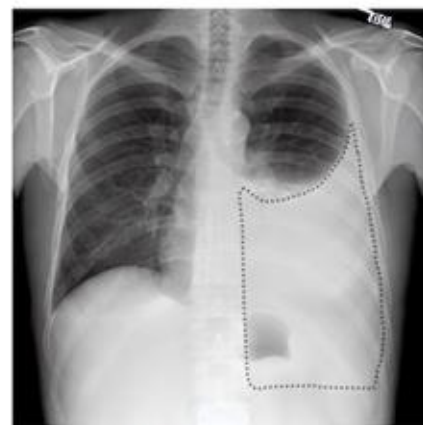
"Small left-sided
pleural effusion"



"Big right-sided
pleural effusion"



"Bilateral
pleural effusions"



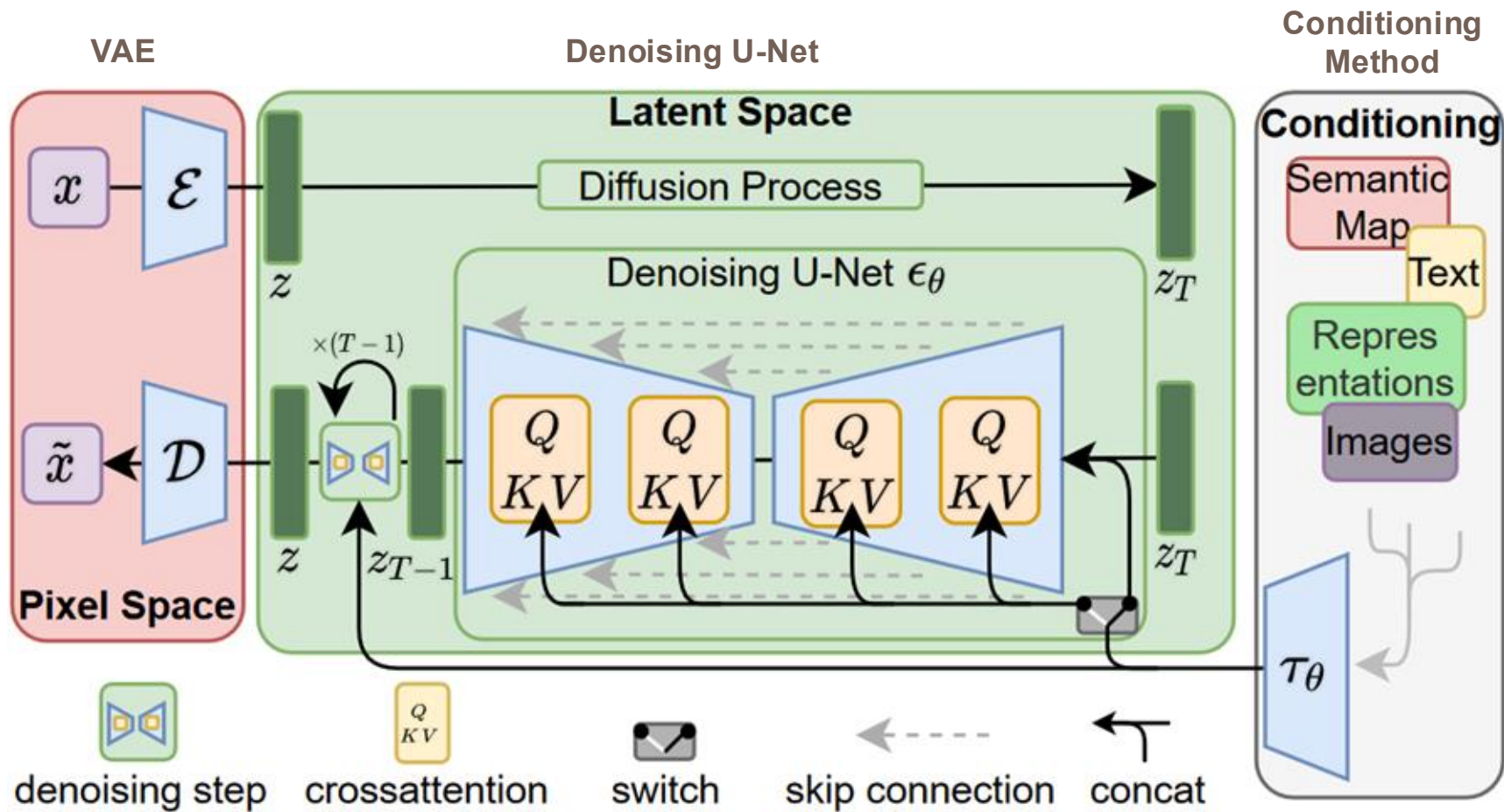
"Big left-sided
pleural effusion"



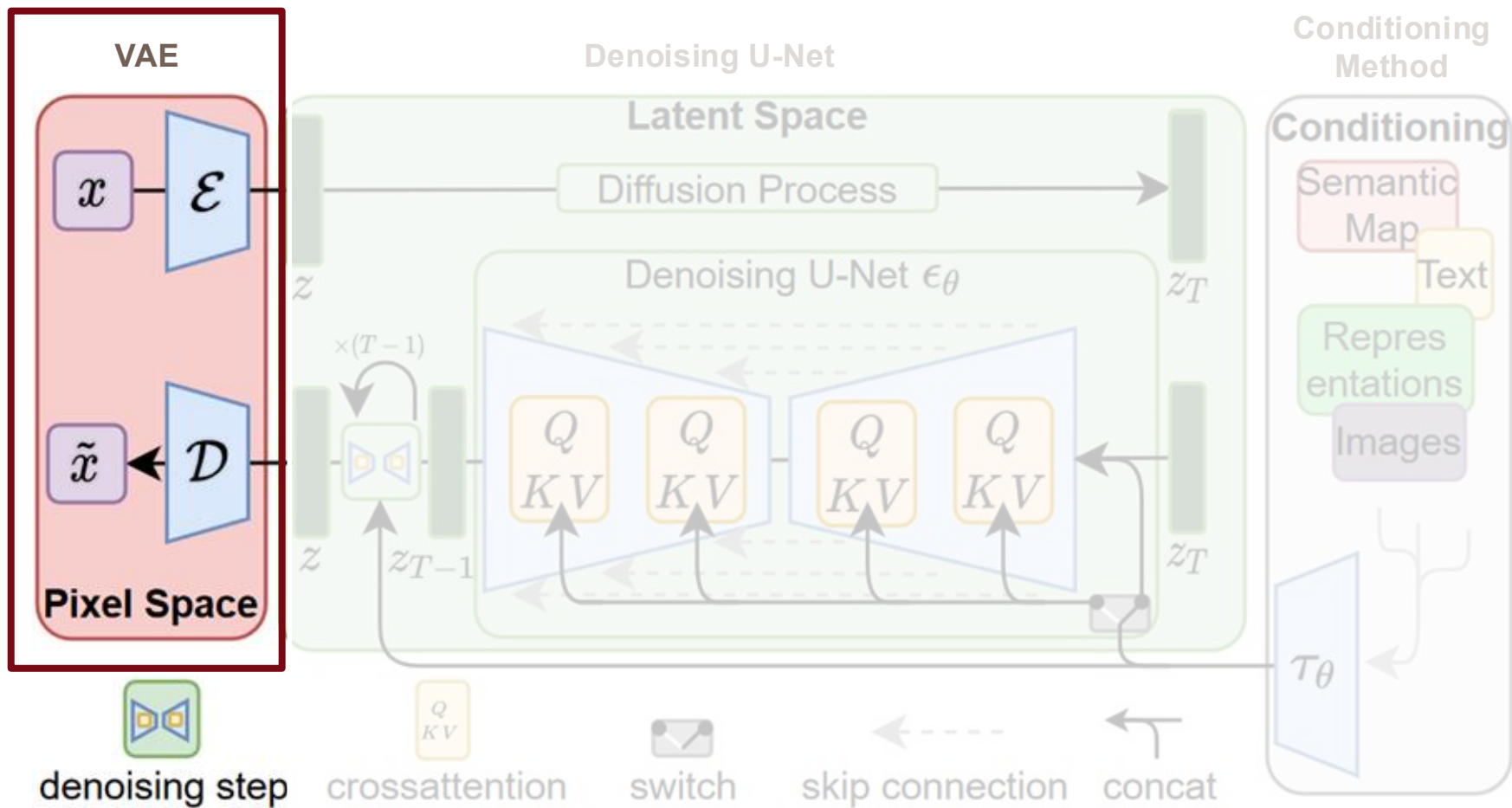
Chris
Bluethgen



Pierre
Chambon

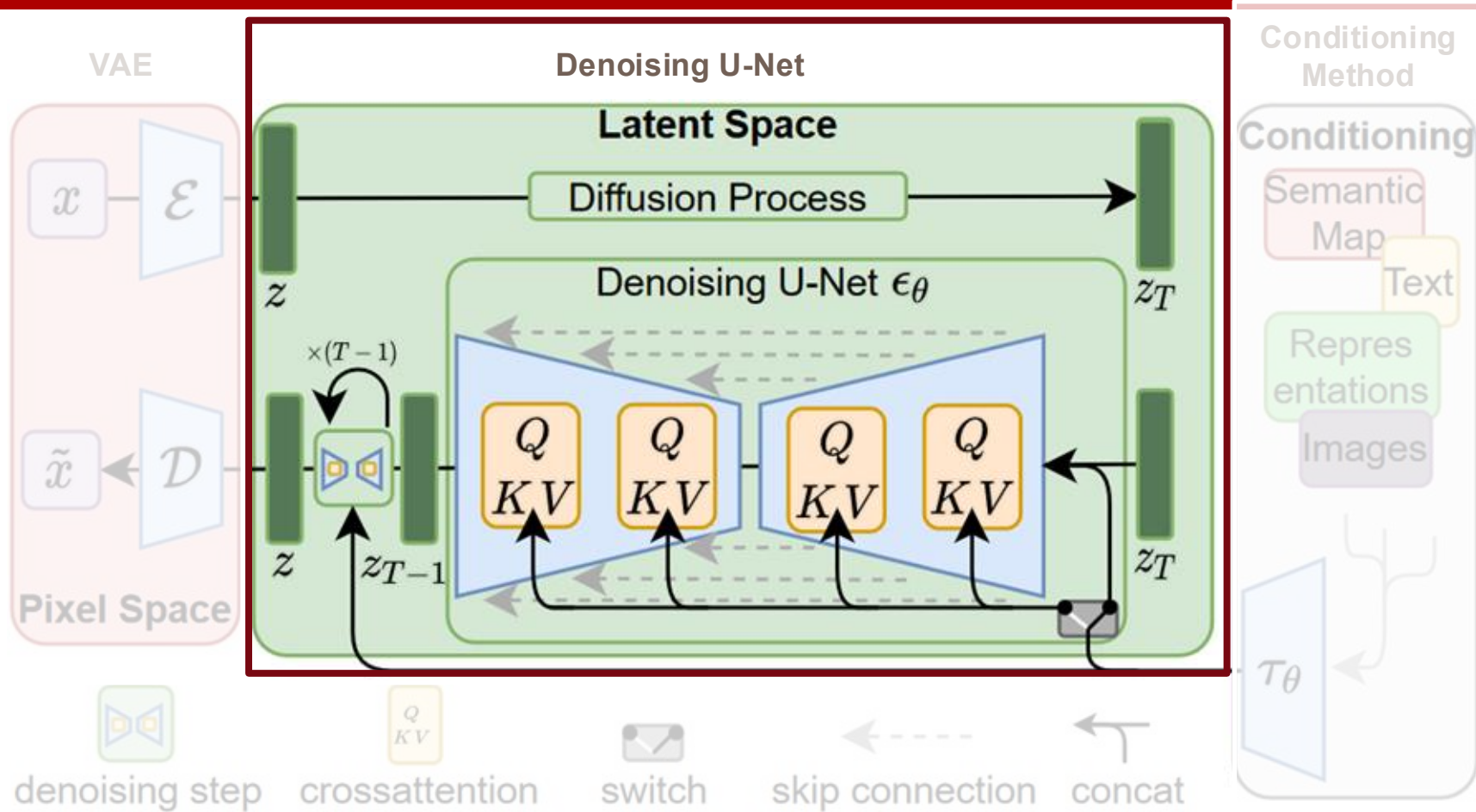


Stable Diffusion

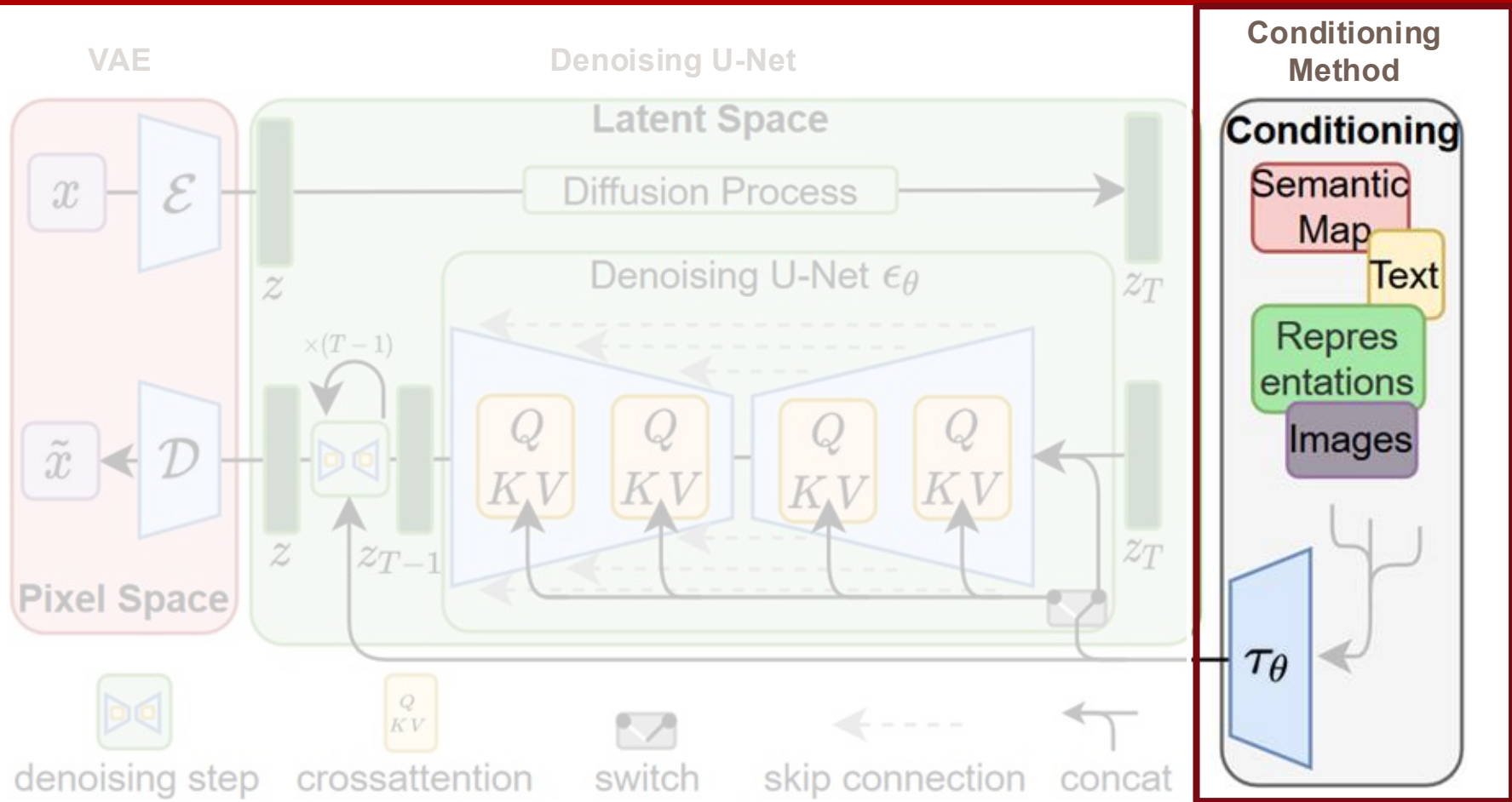


Stable Diffusion





Stable Diffusion



Stable Diffusion

Qualities of (Good) Generative Models

- Fidelity
 - Samples should resemble real data
- Diversity
 - Cover the variety of real data and produce diverse outputs
- Authenticity
 - Samples should not just be a copy of initial data

Fidelity

SD



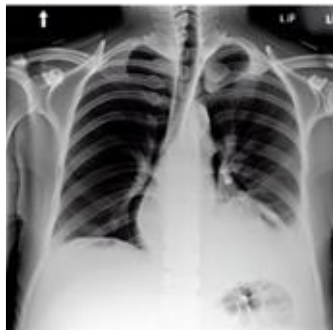
DreamBooth



RoentGen



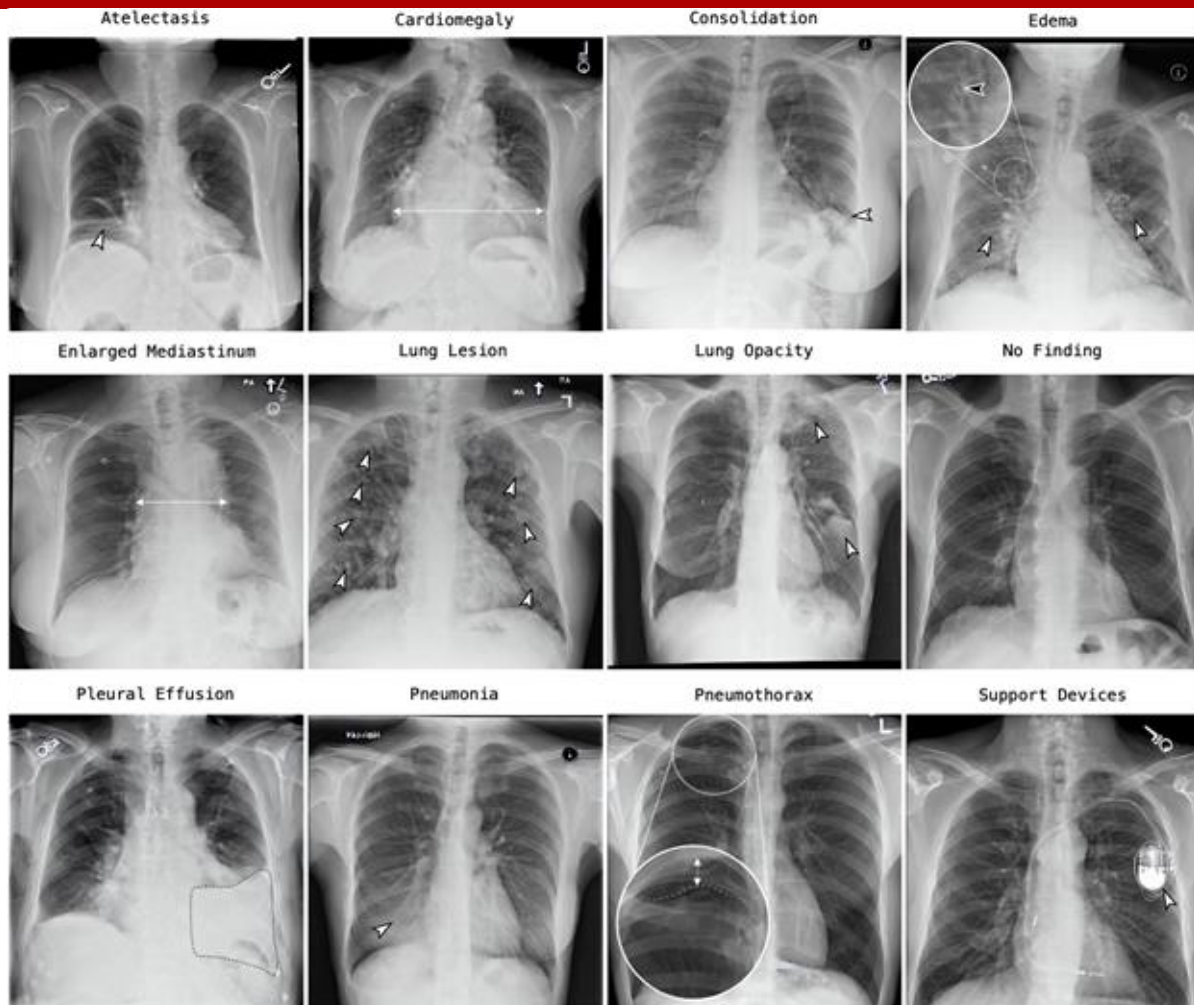
Real CXR



Fidelity

Reference

Diversity

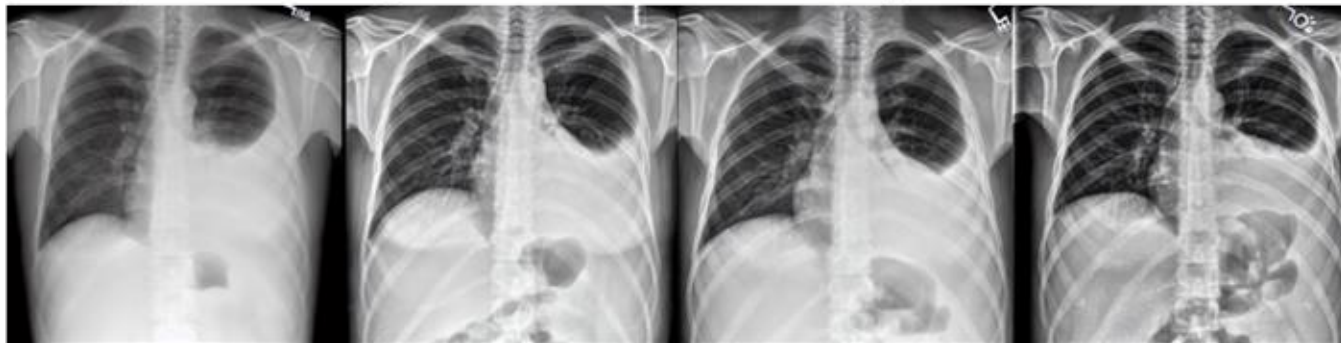


Diversity & Authenticity

"Big right-sided pleural effusion with adjacent atelectasis."



"Big left-sided pleural effusion with adjacent atelectasis."



Benefits of Synthetic Data

